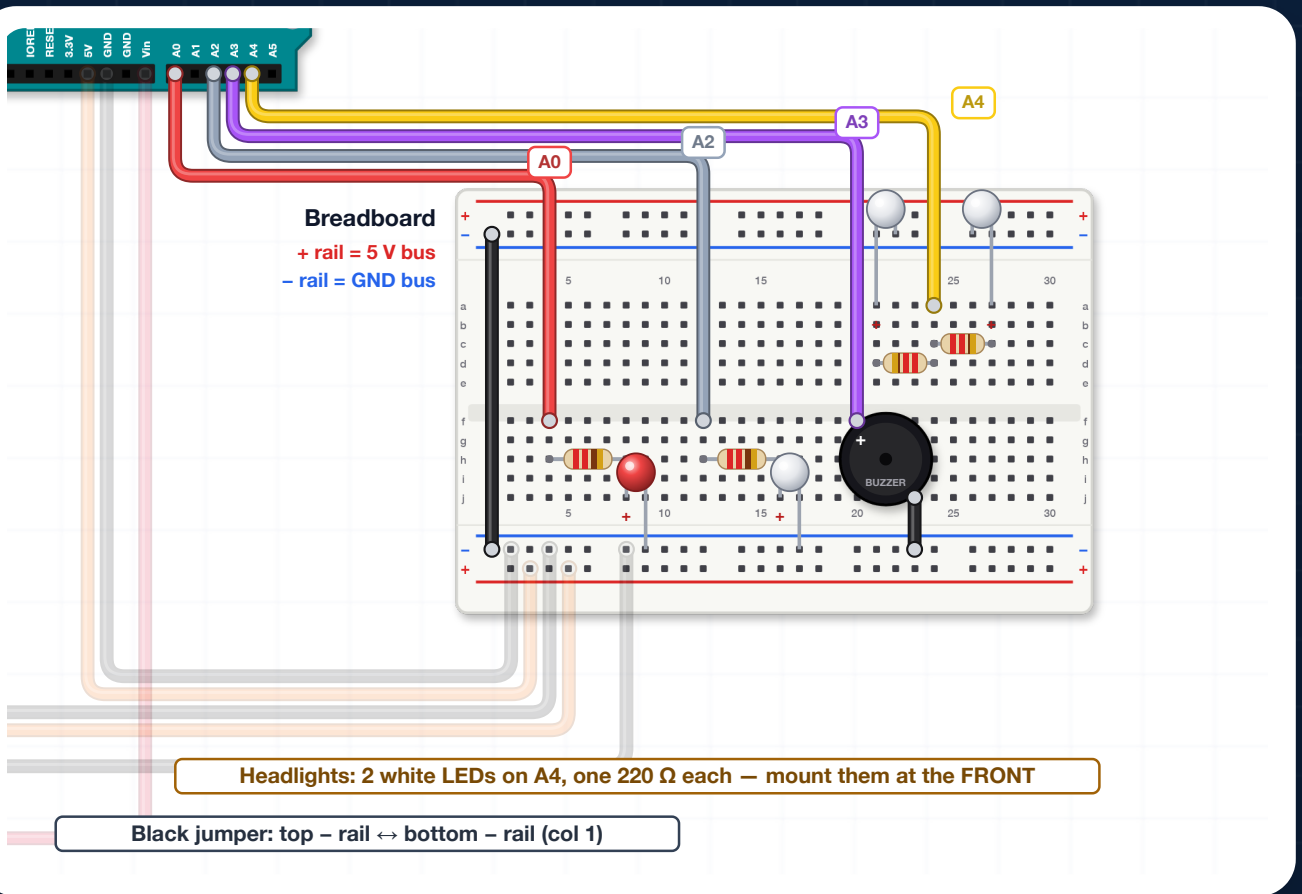
ROBOT CAR PROJECT

MODULE 05

05

Testing & Troubleshooting

Test one part at a time, record the results, and fix the common problems yourself.



Name _____ Class / Group _____ Date _____

4WD · Arduino UNO · ESP32-CAM
Edition 2026

MODULE 05

1 Testing, step by step

Never test everything at once. Upload the small **test sketches** in `tests/` one by one: when something fails, you know exactly which part to look at. Fill in this record as you go.

LEARNING OBJECTIVES · ABOUT 60 MIN

By the end of this module you can:

- Test a system part by part (unit tests before system tests)
- Use the Serial Monitor and a multimeter to find faults
- Fix the most common problems on your own

Lab record: bring-up tests

#	Test	How	Expected result	Pass
1	Power	Battery on, multimeter on the breadboard + and − rails (= UNO 5V)	4.9–5.1 V	☐
2	Motors	Upload `01_motor_test`, wheels in the air	All forward → all back → spin left → spin right	☐
3	Sensor	Upload `02_ultrasonic_test`, Serial Monitor 115200	Distance in cm follows your hand	☐
4	Lights + buzzer	Upload `03_buzzer_led_test`	Headlights on, 4 hazard blinks, then beeps get faster and continuous. Red LED in sync	☐
5	Bench alarm	Upload `06_parking_alarm_bench`, hand towards sensor	SAFE → WARN (faster beeps) → STOP! (continuous)	☐
6	Camera only	Upload `05_esp32cam_test`, join WiFi “CamTest”, open 192.168.4.1	Live video, Photo works, flash LED dims	☐
7	ESP32-CAM	Upload `esp32cam_robot`, Serial Monitor	“Ready.” and WiFi “RobotCar” visible	☐
8	Serial link	Upload `04_link_test` on the UNO, open the app	Radar shows 20→200 cm. Serial shows `M,…` lines	☐
9	Full car	Upload `uno_controller`	Drive with the phone	☐

Acceptance test: the finished car

#	Test	How	Expected result	Pass
1	Drive	Forward 2 m, turn left and right	Smooth, straight, responds immediately	☐
2	Video	Watch the camera while driving	Clear image, little delay	☐
3	Parking alarm	Reverse slowly towards a wall	Beeping starts at about 100 cm and speeds up, continuous near 20 cm	☐

#	Test	How	Expected result	Pass
4	Auto-brake	Keep reversing	Stops at about 15 cm, AUTO-BRAKE tag on the phone. Forward still works	☐
5	Radar	Move a box behind the car	Radar distance and colour change	☐
6	Headlights	Tap ON, then HAZARD in the app	Both front LEDs light. HAZARD blinks them with the red rear LED	☐
7	PC viewer	Open `192.168.4.1/view` on a PC while driving	Video, radar, car status and chart update live	☐
8	Fail-safe	Drive, then turn phone WiFi off	Car stops within 1 s	☐

Tested by (student) _____ Verified by (teacher) _____ Date _____

MODULE 05

2 Troubleshooting

Power & resets

Symptom	Likely cause	Fix
ESP32-CAM reboots when the motors start (`Brownout detector was triggered`)	5 V dips	Check the 2200 µF capacitor (polarity, close to the ESP). ESP on the UNO 5V rail, not the L298N's 5V. Short power wires, good charged 18650s. Try the Slow gear
Everything dies at full speed	Weak or fake batteries	Better cells, charge them, lower the speed slider
L298N gets hot	Normal up to ~60 °C	Keep the heatsink free. Don't stall the motors
UNO regulator (next to the DC jack) very hot	Too much on the 5 V rail	Warm is normal. Never put a servo or motor on the 5 V rail
Nothing turns on with the battery	VIN wire missing or loose	Check the second red wire L298N +12V → UNO VIN and the battery polarity

Motors

Symptom	Fix
Nothing moves, but the app shows L/R values	ENA/ENB jumpers still on? L298N GND connected? Battery on? Chip says “car not responding” → serial link
One side drives backwards	Swap that side's OUT wires, or set `INVERT_LEFT/RIGHT = true`
One motor of a pair fights the other	Swap only that motor's two wires
Turns left when you push right	Swap the left and right OUT pairs (OUT1/2 ↔ OUT3/4)
Hums but doesn't move at low speed	Raise `MIN_PWM` (110–130)
Won't reverse even with nothing behind	Sensor sees the floor or the chassis: mount it higher, tilt it up. Or turn auto-brake off

Camera & WiFi

Symptom	Fix
`Camera init FAILED`	Re-seat the ribbon cable, close the latch. Board = AI Thinker ESP32-CAM
No “RobotCar” network	Check the Serial Monitor. Maybe STA mode is set in `config.h`
Phone switches back to mobile data	Turn mobile data off, or choose “stay connected” when the phone says “no internet”
Video stutters	Settings → QVGA, quality 15–20. Keep the camera away from the motors
Video OK but controls “offline”	Reload the page. Use `http://`, not https. Disable data-saver mode

Sensor & alarm

Symptom	Fix
Radar: “waiting for sensor…”	UNO not sending: check D2 → 1k → IO15, IO14 → D3, common GND, UNO powered
Distance always “>400”	TRIG/ECHO swapped, no 5 V on VCC, or nothing within 4 m
Buzzer silent, LED works	Passive buzzer (needs an active one), wrong polarity, or “Mute the car buzzer” is on

Lights & PC viewer

Symptom	Fix
No headlight lights up	A4 wire in row 24 (top half)? Top − rail bridged to the bottom − rail? Does the LIGHTS tag appear in the app?
Only one headlight works	That LED is backwards (long leg = +), or its resistor is not in the same column as the LED's long leg
PC viewer: “too many viewers”	Max 3 video screens at once. Close other tabs or the photo page
PC viewer: no data, only demo	Open `http://192.168.4.1/view`, not the file from the project folder. PC on the RobotCar WiFi

Multimeter checks

Measure between	Should read
Battery + and −	7.0–8.4 V (recharge below 6.8 V)
UNO 5V (+ rail) and GND	4.9–5.1 V
ESP32-CAM 5V and GND while driving	above 4.75 V
ESP32 IO15 and GND, car idle	≈ 3.3 V (divider OK). **Never 5 V**